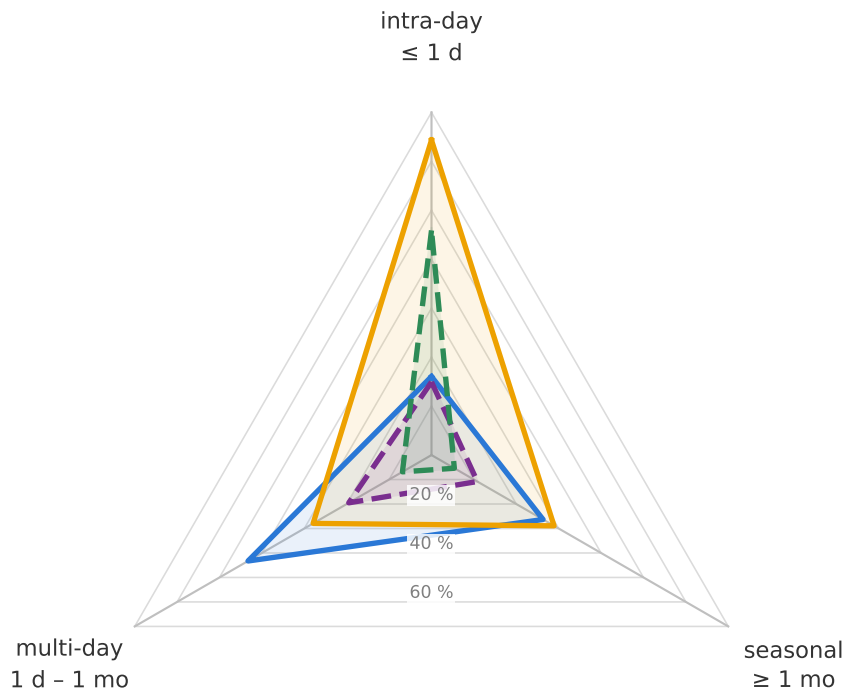


Residual demand variability by period band: pure wind, pure solar, two country mixes
 energy a lossless storage on that band alone would cycle per year, % of annual demand;
 generation = $1.2 \times$ annual demand, no other flexibility; the country cases use
 PyPSA-Earth country-aggregated capacity factors and demand



- wind only, Northern Germany (52.5°N, 10.0°E), mean of 10 years
- solar only, Northern Germany (52.5°N, 10.0°E), mean of 10 years
- Great Britain, 75% wind / 25% solar by energy, 2013
- India, 25% wind / 75% solar by energy, 2013